ICSMMEP-26 — Konuşma Metni

Eggshell–Agar Biocomposites: Development and characterisation for sustainable interior applications Betül Ünal · Cappadocia University · Mecca, 3–4 November 2026

Süre modeli

Metnin kendisi	**13:00** · 140 kelime/dakika
28 slayt geçişi, ekrana dönüp işaret etme, nefes	~1:30
Sahnede beklenen	**≈ 14:30**

Metni bilerek 15:00'in altına yazdım. Sessiz okuduğunda 13 dakika sürer; sahnede hiçbir konuşma kağıt üstündeki süreyle gitmez. Provada 13:00–13:30 çıkıyorsa doğru yoldasın. 14:30'u geçiyorsa hızlanma, en sondaki listeden cümle at.

İngilizce paragraflar konuşulacak metnin kendisidir. Köşeli parantez içindeki Türkçe notlar sahne yönergesidir, okunmaz. Sayıları bilerek yazıyla yazdım; rakam görüp duraksamayasın diye.

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1 · Cover — 0:15

Good morning. Betül Ünal, Department of Interior Architecture, Cappadocia University. The panel in the photograph is the subject of this paper: waste eggshell bound in agar. I will give you the protocol, the characterisation, and an argument about how this class of material should be evaluated.

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2 · Background — 0:35

Eggshell is a continuous food-processing by-product, and it is predominantly landfilled. It is roughly a tenth of the mass of every egg. Chemically it is almost pure calcium carbonate, in a porous protein-bound structure, with a specific gravity below most mineral fillers used in construction.

[Alttaki şemaya işaret et]

So it has two possible fates: landfill, or a secondary mineral resource. The case for eggshell as a mineral phase is settled. This paper is about the phase that binds it.

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3 · Prior applications — 0:35

This is the literature the study departs from. Eggshell has been used as a cement replacement in concrete, a stabiliser in laterite brick and in clay, a filler in asphalt, in eco-ceramic tile, in PVC, in polyurethane foam, in ceramic–metal composites, in bone scaffolds, and in wallpaper.

[Elini orta sütuna götür, aşağı indir]

But read the middle column downwards. In every case the mineral is recovered from waste while the binding phase stays cementitious, ceramic or petrochemical. The environmental burden is carried by the binder, and the binder is untouched.

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4 · Agar as binder — 0:30

The binder here is agar, a thermoreversible polysaccharide from red seaweed. It sets on cooling without chemical crosslinking: no reactive hardener, no cure temperature. Its behaviour with glycerol is well mapped — thirty-two formulations, released as an open dataset.

But note the scope of that work: unfilled thin films, in uniaxial tension, for packaging. The compositional logic transfers. The loading case does not, and for an interior surface that is bending.

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5 · Research gap — 0:35

Three gaps follow. First, the binding phase is never biological: no study reports an eggshell composite in which both phases are biological and fully biodegradable. Second, agar is characterised only as an unfilled film in tension, not as the binder of a mineral-filled cast solid in bending. Third, property sets are fragmentary, in separate studies on separate formulations.

Let me state the contribution precisely. It is not a novel constituent. It is a documented protocol and a single coherent property set.

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6 · Aim — 0:20

The aim, narrow on purpose: document a reproducible protocol for a fully bio-based eggshell–agar composite, and characterise it against published standards. Four objectives follow. The fourth — deriving an evaluation criterion for this material class — is where I will spend most time.

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7 · Scope — 0:20

Scope, set out so that nothing I say is read beyond it. Five formulations, three specimens each, five properties, three standards, ambient cure in one laboratory.

[Alttaki şeride işaret et]

The filled boxes are what was determined. The outlined boxes are what was not: durability, fire classification, life-cycle assessment, structural use.

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8 · Method — 0:20

Four stages, and the order is the method. Produce, recording every parameter. Measure against published standards. Position against two material families. Propose an evaluation criterion derived from the observed behaviour. The properties are measured first and the application envelope is derived from them, not the reverse.

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9 · Research questions — 0:15

Five questions. Two on formulation, two on benchmarking, and the fifth asks whether compressive strength is an adequate criterion for this class at all. One to four are answered by measurement. Five is answered by argument.

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10 · Hypothesis — 0:20

Two clauses. First: the composite can be produced repeatably under ambient conditions. Second: its performance as an interior surface is governed by deformation capacity rather than strength.

On provenance: the dissertation states a broader hypothesis framed on spatial resilience. This is its materials-engineering component, restated within the scope of these tests.

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11 · Material — 0:10

Four constituents: waste eggshell powder, agar, glycerol and water, with a shellac finish. The variation you see is natural pigment and mould geometry, not composition.

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12 · Stage 1 · Mineral phase — 0:30

Stage one. Whole shell from a single bakery, cleaned by hand, heat-treated at one hundred and twenty degrees for twenty minutes, milled, and sieved through a one hundred and fifty micrometre mesh, passing fraction only.

The particle size was chosen to match the eggshell polymer-composite literature, so that the mineral phase here is comparable with that work. And I call the heat step decontamination and drying, not sterilisation, because sterility was not verified.

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13 · Stage 2 · Binder preparation — 0:35

Stage two, drawn as a temperature–time profile. Water and glycerol are pre-mixed before any solids. Agar is added incrementally under continuous stirring; bulk addition forms agglomerates that do not redisperse. The mixture is taken to ninety-five degrees and held until the agar is fully in solution. Then down to eighty while viscosity develops, and the shell goes in near the end; earlier, it sediments. Fifty-five minutes in total.

The set points and total time are recorded; the ramp rates are not, and stirring was manual.

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14 · Stage 3 · Casting and cure — 0:30

Stage three. Day zero, cast into silicone moulds. Demoulded on day four, then seven days at twenty-three degrees, shielded from sun and air movement. Day eleven, finished, with three shellac coats. There is no thermal input after the fifty-five minute preparation: the entire cure is at room temperature.

[Panele işaret et]

The limitation: the composite shrinks as water evaporates, and post-drying dimensions were not logged, so every dimension I quote is a mould dimension.

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15 · Formulations — 0:35

The formulation matrix. Two things to read in it. In the compressive series, shell-to-binder is exactly two point zero zero in all three mixes. That was not planned as a control, but it means solids concentration is the only variable — and you can see it in the bars: the water band shortens from C1 to

C3.

In the flexural series I declare a confound. Glycerol was halved but batch mass was held constant, so water rose to compensate. The plasticiser effect is not cleanly isolated.

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16 · Test programme — 0:20

Four specimen geometries, drawn here with the standard and the number of specimens under each.

I want the reporting convention on record before any numbers appear. Mechanical values are means of three specimens; individual values were not retained, so I cannot give standard deviations. Ignitability, absorption and density are single determinations.

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17 · Compression — 0:35

The compression result, which answers question one. Solids concentration is the only variable, and compressive stress rises linearly with it, from two point five five to four point one zero megapascals, R-squared zero point nine nine.

Three points will fit almost any line, so I call this a hypothesis rather than a law. But the physical reading is clear. Strength is governed by packing density, not by mineral content. The route to a stronger composite is to remove water, not to add shell.

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18 · Compressive failure — 0:35

This is the failure mode. Linear to about one megapascal, then progressive loss of stiffness as the agar network yields and the mineral redistributes. Peak near ten per cent strain, no abrupt load drop, and beyond peak it keeps deforming while holding together.

[Fotoğrafı göster]

The cube has barrelled. It is shorter and wider, and still a single body. Failure proceeds by progressive crushing rather than by fracture. One honesty point: the strain axis assumes ten per cent maximum deformation from observation, not instrumented measurement.

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19 · Flexure — 0:35

Flexure, to ASTM D790. The decisive variable is plasticiser content. At eighteen point eight per cent glycerol the composite deformed without developing measurable resistance. Halving it gives two point three two five megapascals at a modulus of one hundred and ninety-two. The transition is sharp, and it lies between those two values.

Two limitations. The glycerol effect is confounded with solids concentration. And the specimen is kerfed on its tension face while the standard assumes a solid rectangle, so this modulus is a lower bound.

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20 · Flexural deformation — 0:20

I would rather show you this than ask you to take it on the numbers. The beam is deflected well past where a comparable ceramic would fail, and it has not parted. Both test programmes gave the same result: substantial deformation with retained integrity.

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21 · Fire, water, density — 0:35

Ignition at around ten seconds under direct flame, self-extinguishing four seconds after removal, with no lateral spread and no dripping. The probable mechanism is endothermic decomposition of the carbonate fraction, diluting the flame front; not verified calorimetrically. This is an ignitability test. There is no Euroclass classification.

Water: zero point seven per cent mass gain over twenty-four hours of immersion, so coating integrity is the limiting factor, not the composite. Density, one point three eight: above the fungal materials, below fired clay.

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22 · Conventional materials — 0:30

First benchmark, against what this would sit alongside in an interior. Let me be direct. One sixth the compressive strength of MDF, one twentieth that of travertine — and travertine is the relevant aesthetic comparison, because the material resembles it. On strength alone this composite is not competitive with engineered wood or stone, and I will not claim otherwise.

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23 · Bio-based composites — 0:35

Second benchmark, and this is the one that matters. Against the bio-based family the position changes completely. In compression, an order of magnitude above mycelium and hempcrete.

[Logaritmik eksene işaret et]

In stiffness, one hundred and ninety-two megapascals: inside the LDPE band. Note what that means in process terms. Water-cast, dried at room temperature, no oven, no autoclave, no extruder, and it reaches the stiffness of melt-processed bio-polymers. Wood is still two orders of magnitude above.

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24 · Modulus of resilience — 0:55

This brings me to question five, and the diagram is the argument.

The shaded area is the energy absorbed per unit volume before permanent damage. That area is one half sigma epsilon, and epsilon is sigma over E, so it reduces to sigma squared over two E. Notice where the modulus sits: in the denominator. The low modulus that counted against the material a moment ago is what raises this quantity. For this formulation it is about fourteen kilojoules per cubic metre.

Strength records the load at which a material stops performing, not whether it then fails gracefully or catastrophically — and for a demountable non-loadbearing component, that is what governs service life.

The limitation: I cannot give you a comparative table, because the literature reports strength and modulus separately. This is a reporting convention, not a ranking.

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25 · Application envelope — 0:30

The envelope follows from the measurements. Non-loadbearing, because four point one megapascals supports nothing more. Dry interiors, because water performance depends on the coating. Away from ignition sources: self-extinguishing, but unclassified. Modular rather than sheet, because damage is then replaced one unit at a time.

Every constituent including the shellac is biodegradable. No life-cycle assessment was carried out, so these are qualitative potentials, not quantified claims.

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26 · Limitations and further work — 0:35

The limitations, each next to the experiment that resolves it, because that pairing is the point. No dispersion data, because individual values were not retained — repeat the series and retain them. Strain assumed from observation — instrument it. Shrinkage unquantified — record the post-drying geometry. The kerfed section calculated as solid — test an un-kerfed one. The flexural series confounded — vary glycerol at constant solids concentration.

Every value I reported today is a lower bound. I would rather set these out than have them found.

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27 · Conclusions — 0:30

Five conclusions. A protocol that can be repeated, parameter by parameter. Packing density, not mineral content, governs compressive strength. Plasticiser content sets a brittle or a ductile response, and the transition is sharp. A water-cast, ambient-dried material reaches bio-polymer stiffness without thermal processing. And strength alone is an insufficient criterion for this class: deformation capacity governs service behaviour, so strength and modulus should be reported together.

Thank you. I am happy to take questions.

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28 · References — 0:00

[Slaytı açık bırak, bir şey söyleme. Soru gelirse kaynağı buradan göster.]

Soru–cevap

Kısa tut. Sınırlılığı kabul et, sonraki deneyi söyle, sus.

"Standart sapma yok, nasıl güveneceğiz?" You are right, and I said so before the numbers. Three specimens per formulation, individual values not retained. Repeating the series with dispersion reported is the first row of my limitations table.

"Basma testi hangi standarda göre?" Cube compression, not to a named standard, and I flag that as a deviation. Flexure follows ASTM D790-17, ignitability ISO 11925-2, absorption ASTM D570.

"Ölçüler kalıp ölçüsü, gerçek gerilme değil." Correct. The composite shrinks as water evaporates and post-drying dimensions were not logged, so every value I reported is a lower bound. Recording the dried geometry is in the limitations table.

"Yivli numuneden modül hesaplanmaz." Agreed, which is why I called it a lower bound rather than a material constant. Testing an un-kerfed section is in the table.

"R^2 = 0.99 ama üç nokta var." That is exactly why I called it a hypothesis rather than a law. What I claim is the physical reading: packing density, not mineral content, governs strength.

"Dirençlilik modülü için karşılaştırma yok." No, and I cannot assemble one, because the literature reports strength and modulus in separate studies on separate formulations. I am proposing a reporting convention, not a ranking.

"Nem, küf, yaşlanma?" Untested, and named as such. Durability of any kind is outside the scope of this paper.

"Ölçeklenebilir mi?" I have no data on that. What I can say is that the protocol needs a hotplate, a mould and ambient drying. No oven, no autoclave, no extruder.

"Neden agar? Nişasta ya da jelatin olmaz mıydı?" Thermoreversible setting without a hardener, and it tolerated sixty-six per cent mineral loading without a coupling agent. Comparing binders is a separate study, and a fair one to ask for.

"Yangın sınıflandırması olmadan iç mekanda nasıl kullanılacak?" It cannot be specified without one, and I say so on the application slide. What I report is ignitability behaviour, which is the precondition for seeking a classification, not a substitute for it.

Provaya dair

- Yüksek sesle, kronometreyle oku. Metnin kendisi 13:00; sahnede duraklarla

birlikte 14:30 civarı. 13:00–13:30 çıkıyorsa doğru hızdasın.

- :30'u geçersen **hızlanarak değil, cümle atarak kısalt.** Sırasıyla:
- Slayt 16, ikinci paragrafın son cümlesi (*"Ignitability, absorption and

density are single determinations."*)

- Slayt 22, *"and travertine is the relevant aesthetic comparison…"* ara

cümlesi.

- Slayt 25, son paragraf.
- Slayt 7, *"The outlined boxes are what was not…"* — ekranda zaten yazıyor,

yalnız işaret et.

- Slayt 12, ikinci paragraf.
- Dokunulmayacak slaytlar: **3, 5, 13, 15, 17, 19, 24.** İskelet bunlar.
- Slayt 24 sunumun tepe noktası. Oraya kadar tempoyu koru, orada yavaşla.
- Köşeli parantezli yönergelerde gerçekten ekrana dön ve işaret et. O üç saniyelik

duraklar hem ritmi kuruyor hem de süre modelindeki 1:30'un bir kısmı.